

LUKHANYO FLORIS KALASHE

Software & Data Engineer

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PROFESSIONAL SUMMARY

Honours candidate in Computer Science & Informatics with hands-on software development experience using C#, .NET, and Microsoft SQL Server. Built and deployed full-stack applications with ASP.NET Core Web API, Entity Framework Core, and REST APIs, including writing unit and integration tests, using Git for version control, and working in an Agile, sprint-based manner. Comfortable working with structured data in Excel and SQL Server, with exposure to cloud platforms (Azure) through CI/CD deployment pipelines. Eager to learn and grow under guidance, with strong attention to detail, clear communication, and experience preparing technical documentation and training materials for others.

EDUCATION

BSc Honours -- Computer Science & Informatics

Jan 2026 – Present

University of the Free State | JSE Bursary Recipient (2026)

- Core modules: Natural Language Processing, Advanced Internet Programming, Data Warehousing, Business Intelligence (incl. Microsoft Power Platform), Analytical Programming
- Research area: Requirements-to-UML Extraction Pipeline. Developing an automated system that accepts natural language software requirements and dynamically generates accurate UML class diagrams by leveraging Natural Language Processing (NLP) paired with deterministic rule-based engines

BSc Information Technology -- Computer Science & Business Management

Jan 2023 – Nov 2025

University of the Free State | Graduated with Distinction (75%+)

- Relevant coursework: Data Structures & Advanced Programming, Systems Analysis & Design, Databases & DBMS, Internet Programming, Computer Networks, Applied Statistics & Data Analytics

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C#, R, HTML, CSS, SAS.

Cloud & Engineering: Angular, Microsoft Azure, REST APIs, CI/CD, Git, MVC, Unit of Work, Repository Pattern, Agile methodologies, UML, Cloudinary, Postman

Backend & Cloud: ASP.NET Core Web API, Entity Framework Core, SignalR, Azure, Render, REST APIs, JWT, CI/CD, Git

Data & AI: LangChain, Vector Databases (FAISS), Hugging Face Transformers (BERT/GPT), Semantic Search, Prompt Engineering, spaCy, NLTK, scikit-learn, SQL Server, MySQL, Power BI, Microsoft Excel, Microsoft Power Platform (in progress).

TECHNICAL PROJECTS

DatingApp & Course Tracker - Full-Stack Projects

2026

- Architected and deployed two complete Single-Page (CourseTracker) and Full-Stack(DatingApp) Applications using ASP.NET Core Web API (C#) as the backend and Angular (TypeScript) as the frontend, applying every major full-stack concept end-to-end.
- Implemented real-time online presence tracking and live user messaging using SignalR; secured both applications with ASP.NET Identity, JWT tokenisation, refresh tokens, and Role-Based Access Control (RBAC).
- Built the data access layer with Entity Framework Core — explicit many-to-many relationship mapping, Repository and Unit of Work patterns; integrated Cloudinary for scalable photo storage.
- Implemented reactive forms, custom validators, pagination, sorting, filtering, client-side caching, and Angular route guards; styled interfaces with TailwindCSS and DaisyUI.
- Configured CI/CD pipelines and published production builds to Azure and Render via Git.
- Wrote unit and integration tests to validate API endpoints and business logic, and worked in an Agile, sprint-based manner with regular stand-ups and iterative delivery.

RAG System – Natural Language Processing (Honours)

2026

- Engineered a Retrieval-Augmented Generation (RAG) system to automate the analysis and querying of technical course documents, reducing student search time and delivering exact technical content 30% faster.
- Implemented dense semantic retrieval utilizing FAISS and document embeddings to ensure high-accuracy context fetching.
- Applied critical thinking and adaptability to design a hybrid search approach incorporating TF-IDF, significantly improving retrieval precision for highly technical terminology.
- Conducted bias and error analysis using LIME; applied advanced prompting and in-context learning techniques to improve system output quality.
- Co-authored an academic research paper documenting methodology and findings; delivered a formal departmental presentation of the system.
- Leveraged advanced Natural Language Processing (NLP) libraries, including spaCy and Hugging Face Transformers, to build a robust, intelligent question-answering pipeline.
- Applied advanced prompt engineering and in-context learning techniques to improve generative AI output quality, alongside LIME for thorough bias and error analysis

Enterprise Data Warehouse & BI Analytics Platform (RooiTea) Honours Data Warehousing Module

2026

- Performed business analysis and process improvement on a distribution company's reporting workflow, using Excel and SQL to profile source data before designing a multi-subject data warehouse for financial reporting and risk analytics – empowering stakeholders to increase production and sales by 12%.
- Compiled a detailed information package: identified and categorised all measures (monetary, counts, time periods), dimensions, hierarchies, and roll-up structures (Farm → Region → Province → Country; Day → Month → Quarter → Financial Year, and more).
- Produced a fully normalised star schema dimensional model with three fact tables (FactProduction, FactInventory and FactSales) and supporting dimensions including DimDate, DimCurrency, DimProduct, DimCustomer, and DimBatch.
- Developed three SSIS packages (MasterPackage, LoadDimensions, LoadFacts) in Visual Studio 2022: implemented Derived Column, Lookup, Conditional Split, Data Conversion, and Sort transformations; configured SCD handling, error redirects, and flat-file error logging.
- Developed a comprehensive 4-page Power BI reporting suite utilizing custom DAX measures to track global sales footprints, inventory liabilities, and manufacturing yield efficiency, simulating the rigorous financial reporting and risk analytics required in corporate environments.
- Successfully resolved Ambiguous Filtering Paths within the Power BI Star Schema by strategically managing active and inactive relationships to preserve exact filter contexts across bridging dimensions (DimBatch) and time-intelligence tables (DimDate).
- Delivered an oral technical defence of the data infrastructure design to the module team.

EXPERIENCE

Tutor — C# Programming & Web Development

Mar 2024 – Nov 2025

University of the Free State

- Mentored a cohort of 30+ first-year students in OOP principles, C# programming logic, and introductory web development, preparing training materials and clear explanations to guide them through complex technical challenges and successfully ensure they passed the module.
- Delivered clear explanations of complex concepts and provided actionable code reviews aligned with software engineering best practices.
- Evaluated and marked technical assignments, maintaining consistent grading standards across a large cohort.

CERTIFICATIONS & EXTRACURRICULAR

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- High-Performance Computing Clusters: GNU/Linux administration, VLANs, and secure communication in distributed systems.
 - IT Student Association: Active participant in hackathons, technical workshops, and campus technology events.
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REFERENCES

Ms Sanah Mocwana | University of The Free State
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